Mini Capstone Project Proposal

Title

Employee Attrition Prediction System using Machine Learning

Problem Statement

Employee attrition is one of the major challenges faced by organizations as it increases recruitment costs, reduces productivity, and results in the loss of experienced employees. HR departments need efficient methods to identify employees who are likely to leave so that timely retention strategies can be implemented.

This project develops a Machine Learning-based Employee Attrition Prediction System that predicts whether an employee is likely to stay or leave based on factors such as Age, Department, Job Role, Monthly Income, Years at Company, Overtime, Job Satisfaction, and Work-Life Balance. The system compares Logistic Regression, Decision Tree, and Random Forest algorithms and provides attrition probability through an interactive Streamlit dashboard.

Aim

To develop a Machine Learning web application that predicts employee attrition and compares multiple classification algorithms.

Objectives

- Predict employee attrition.
- Compare Logistic Regression, Decision Tree, and Random Forest or any 3.
- Display attrition probability.
- Provide HR analytics dashboard.
- Assist HR in retention decisions.

Dataset

IBM HR Analytics Employee Attrition & Performance Dataset (Free for educational use).

Feature	Type
Age	Integer
Department	Categorical
JobRole	Categorical
MonthlyIncome	Integer

OverTime	Yes/No
JobSatisfaction	1-4
WorkLifeBalance	1-4
YearsAtCompany	Integer
Attrition	Target

Algorithms

- Logistic Regression
- Decision Tree
- Random Forest
- Or any 3 as per your choice.

Expected Output

Employee Will Stay / Employee Will Leave
Attrition Probability
Best Performing Algorithm

Technology Stack

Python, Streamlit, Pandas, NumPy, Scikit-learn, Plotly, Matplotlib, Joblib

Project Workflow

Dataset → Preprocessing → EDA → Feature Encoding → Train/Test Split → Train Models → Evaluate → Compare → Streamlit App → Prediction